Tools kont use kel:

“I used Python as the programming language, along with PyTorch for building the CNN model, OpenCV for image and video processing, and NumPy for numerical operations.”

Preprocessing ks keli:

“Preprocessing मध्ये मी सर्व images 256×256 size मध्ये resize केल्या जेणेकरून सगळ्या images एकसारख्या राहतील. त्यानंतर pixel values 0 ते 1 मध्ये normalize केल्या, ज्यामुळे model चं learning better होतं. मग image format HWC वरून CHW मध्ये बदलला आणि शेवटी PyTorch साठी tensor format मध्ये convert केलं.”

Why preprocessing is important?

“It standardizes the input data and improves model performance.”

Height estimate kse kel

“I estimated the height using a simple pixel scaling method. First, I calculated the height of the detected object in pixels using the bounding box. Then I multiplied it by a scale factor to convert it into real-world height.”

“मी height estimation pixel scaling method वापरून केली. आधी bounding box मधून object ची pixel height काढली आणि नंतर ती scale factor ने multiply करून real-world height estimate केली.”

Annotations manje kay ani tu ka kel:

“Annotation is the process of labeling images by marking the object location using bounding boxes and assigning class labels.”

“Annotation म्हणजे image मध्ये object कुठे आहे हे bounding box वापरून mark करणे आणि त्याला class label देणे.”

Why:

“Annotation is required to train the model because it tells the model where the object is located in the image.”

“Annotation model training साठी आवश्यक आहे कारण त्यातून model ला object कुठे आहे हे समजतं.”

👉 Model ला स्वतःहून कळत नाही:

❌ “object कुठे आहे”
❌ “object कोणता आहे”

👉 म्हणून:
✔️ आपण manually सांगतो

➡️ ते म्हणजे annotation

What is optimizer and which one did you use?”:

“An optimizer is an algorithm that updates the model’s weights to reduce the loss and improve performance during training.”

“Optimizer हा असा algorithm आहे जो training दरम्यान model चे weights update करून error (loss) कमी करतो.

“I used the Adam optimizer.”

“Adam is fast, efficient, and automatically adjusts the learning rate during training.”

Epoch manje kay ani tu use keles ani ka kiti ghetle

“An epoch means one complete pass of the entire training dataset through the model.”

“Epoch म्हणजे पूर्ण dataset एकदा model मधून pass होणे.”

“I trained the model for 30 epochs.”

“I chose 30 epochs because it allowed the model to learn properly without overfitting.”

What is batches

“A batch is a small subset of the dataset used to train the model in one iteration.”

Batch म्हणजे dataset मधील छोटा भाग जो एकावेळी model ला training साठी दिला जातो.”

👉 समज:

- Total images = 100

👉 जर batch size = 10 असेल तर:

100 images ÷ 10 = 10 batches

👉 म्हणजे:

- model एकावेळी 10 images शिकतो

“I used a batch size of 8.”

“मी batch size 8 घेतली कारण ती memory आणि training stability मध्ये balance ठेवते.”

👉 छोटा batch (like 8):
✔️ less memory
✔️ better learning

Concept Meaning

Epoch	Full dataset
Batch	Small part

"Output video format मध्ये दाखवण्यासाठी तू kay use kels

- **English:**

"To generate the output in video format, I used OpenCV for reading the input video, processing each frame, drawing bounding boxes, and displaying the output. I also used VideoCapture to read the video and imshow to display the processed frames."

- Marathi:

"Output video format साठी मी OpenCV वापरलं. त्यात VideoCapture वापरून video read केला, प्रत्येक frame process केला, bounding box draw केला आणि imshow वापरून output screen वर दाखवला."

Bounding box म्हणजे काय? आणि तू ते कसं व का draw केलं?":

- Answer (English)

"A bounding box is a rectangular box drawn around an object to show its location in an image."

- Answer (Marathi)

"Bounding box म्हणजे image मध्ये object कुठे आहे हे दाखवण्यासाठी त्याच्या भोवती काढलेला चौकोन (rectangle) आहे."

Why :

Bounding box is drawn to visually highlight the detected object and show its exact location."

🧠 **Library-wise Explanation:**

1. import os:

   "The os library is used to handle file paths and access folders."

   "os library files आणि folders handle करण्यासाठी वापरली."

   🎯 **का वापरली?**

   ✔ images आणि labels read करण्यासाठी

   ✔ correct file path तयार करण्यासाठी

import cv2 (OpenCV):

"OpenCV is used for image and video processing."

🎯 **का वापरली?**

✔ image load करायला

✔ resize करायला

✔ video process करायला

✔ bounding box draw करायला

3. import torch

"PyTorch is used for building and training the deep learning model."

4. import numpy as np:

"NumPy is used for numerical operations and array handling."

🎯 का वापरली?

✔ image format change

✔ data process

5.import torch.nn as nn:

"torch.nn is used to define neural network layers."

Why:

✔ CNN model बनवण्यासाठी
✔ loss function define करण्यासाठी

🔹 **6. import torch.optim as optim:**

"torch.optim is used to define the optimizer."

Why:

✔ model weights update करायला
✔ loss कमी करायला

🔹 **7. Dataset, DataLoader, random_split:**

"These are used to load and manage data efficiently."

का वापरली?

✔ **custom dataset तयार करायला**
✔ **batch मध्ये data द्यायला**
✔ **train/test split करायला**

❓ **Q8: What is CNN?**

👉 Answer:

"CNN is a deep learning model used for image processing that extracts features using convolution layers."

❓ **Q10: Which loss function did you use?**

👉 Answer:

"I used SmoothL1Loss because it works well for bounding box regression."

❓ **Q12: Did you split the dataset?**

👉 Answer:

“Yes, I used an 80-20 split for training and testing.”

❓ **Q13: How does your detection work?**

👉 Answer:

“The model processes each video frame, predicts bounding box coordinates, and draws a rectangle around the detected object.”

❓ **Q16: Is the height accurate?**

👉 Answer:

“It is approximate and depends on the scale factor and camera angle.”

❓ **Q17: How did you evaluate your model?**

👉 Answer:

“I used training and testing loss, and also implemented a custom accuracy metric.”

❓ **Q19: What challenges did you face?**

👉 Answer:

“Dataset mismatch, incorrect labels, and model not detecting objects initially.”

❓ **Q20: How did you solve them?**

👉 Answer:

“I fixed label naming issues, filtered invalid data, and improved model architecture.”

❓ **Q21: How can you improve this project?**

👉 Answer:

“By increasing dataset size, using better models, and improving height estimation methods.”

❓ **Q22: Why your loss became zero?**

👉 Answer:

“It could be due to overfitting or very small dataset.”

❓ **Q23: What is overfitting?**

👉 Answer:

“When model performs well on training data but poorly on new data.”

❓ **Q24: What will you do if detection fails?**

👉 Answer:

“I will improve dataset quality and retrain the model.”

👉 Interview end ला बोल:

“This project helped me understand the complete pipeline of object detection, from data preparation to model training and real-time deployment.”